

The Sun's five-minute oscillations observed with an amateur spectroheliograph

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Abstract. The surface of the Sun vibrates: it rises and falls with a period close to five minutes. We show here that these vibrations can be measured from the ground with an amateur spectroheliograph and a 102 mm refractor. On 15 September 2026, 336 scans of the solar disc were recorded over six hours in a line of iron. The JSol'Ex software reconstructs the images and measures, at every point of the surface, the velocity of the material along the line of sight. By sorting these motions according to their rhythm and their size, we obtain a diagram in which the power is concentrated along narrow lines: each one corresponds to a family of vibrations, and this is the expected signature of sound waves trapped beneath the surface. Six of these lines are separated, and their frequencies are compatible with those measured from space by the SOHO satellite. This detection and identification do not depend on any instrumental correction. An imperfection in the mount's tracking, which slightly shifts the content of the images, is then measured and corrected: the lines become sharper, while their frequencies do not change. Another family of waves, called the f mode, is also detected, at the position predicted by theory, and its frequency recovers the mass of the Sun, which checks the measurement as a whole. Finally, we describe a pitfall met in a first version of this work: if the analysed area does not follow the surface of the Sun exactly, every line splits in two, and a spurious feature appears.

1 Introduction

In 1962, Leighton, Noyes and Simon [1] discovered that the surface of the Sun oscillates, with a period of about five minutes. They measured this motion through the Doppler effect: when a region of the surface moves towards us, the light it emits is slightly shifted towards blue, and towards red when it moves away.

Ulrich [2] explained these oscillations: they are sound waves, trapped in the layers beneath the surface, which resonate like sound in a musical instrument. An important consequence of this explanation is that not every combination of rhythm and size is possible. Only some are, just as an organ pipe produces only certain notes. If the motions of the surface are sorted according to their rhythm, that is their frequency, and according to their size, that is the distance between two crests of the wave, the power is therefore not distributed at random: it is concentrated along narrow lines, called ridges. Each ridge corresponds to a family of waves that probe the interior of the Sun at different characteristic depths.

Deubner [3] observed these lines for the first time in 1975. He could distinguish three or four of them. Space observations then went much further, with the SOHO satellite [4], and later with Hinode [5].

From the ground, turbulence in the Earth's atmosphere blurs fine details and limits what can be seen [6]. Amateur spectroheliographs, derived from the Sol'Ex, record the complete shape of a line of the solar spectrum at every point of the disc, with a quality recognised by professional observatories [7]. They therefore give access to the velocity of the material at every point. The question asked here is simple: can such an instrument, on an amateur mount

and under an ordinary sky, recover Deubner's ridges and identify them by comparison with space measurements?

One clarification before going further: these oscillations can be seen in two ways. The first is the variation in the brightness of the surface. It requires no more than a sequence of images and careful photometry: three hours of white-light images taken with a 60 mm refractor were enough to produce a diagram of this kind [8]. The second is the measurement of the velocity itself, through the Doppler effect: the shape of the line must then be recorded at every point, a reference is needed for each pixel, and the drifts of the instrument have to be kept under control over several hours. It is this second, more demanding route that is followed here; it is also the route of the space measurements against which we will compare our results.

Two quantities are enough to read the diagram. The first is the **frequency**, expressed in millihertz: 3 mHz corresponds to one oscillation every 333 seconds, a little more than five minutes. The second is the **degree**, written ℓ : it is the harmonic degree, which characterises the size of the structures at the surface: the higher it is, the smaller the waves. At degree 800, a complete wave measures about 5,500 km, that is 2,700 km from a crest to the neighbouring trough, to be compared with the solar radius of 696,000 km. This correspondence between size and degree is that of Gizon and Birch [9].

2 Instrument and observations

2.1 Instrument

Table 1: Instrument.

Item	Characteristic
Refractor	SVBony SV48P, 102 mm, 663 mm focal length
Spectroheliograph	MLAstro SHG 700, 7 μ m slit
Camera	G3M678M, 1 $\times$ 1 binning
Mount	MLAstro SAL-33, with harmonic drives
Size of a pixel on the Sun	0.69 arcsecond, i.e. 503 km at disc centre
Dispersion	one pixel of the spectrum corresponds to a velocity of 5,737 m/s

As an indication, this equipment costs about 2,700 USD at the list prices of September 2026: 275 USD for the refractor, 1,140 USD for the spectroheliograph supplied with its camera, and 1,249 USD for the mount¹. This total includes neither the tripod, nor the power supply, nor the acquisition computer. It is worth noting that this equipment can be reused for other experiments, in spectrography as well as in conventional observing: its cost is therefore fairly easy to recoup.

The line chosen is that of neutral iron at 5434.5 Å. Its effective Landé factor is very small, which makes it only weakly sensitive to magnetic fields under ordinary photospheric conditions and limits the risk of mistaking a magnetic effect for a motion; Zeeman effects nevertheless remain measurable in the strongest fields, at the core of sunspots [10]. It has already been used to study waves at the solar surface [11]. It is wide enough to be sampled by more than three pixels of the spectrum, which is necessary to measure its position accurately.

A word about the height at which this line forms, since we claim to measure motions of the photosphere. The core of the line forms about 550 km above the level where the atmosphere becomes opaque, that is in the upper photosphere, and the line carries information about the whole thickness of it [12]. The nine images used here are taken on either side of the core, and the wings of the line form lower down: the velocity obtained is therefore an average over a slice of atmosphere. Only the strongest lines, such as the hydrogen H α line, form in the chromosphere, higher up.

The choice of this line is first of all a matter of comparison: it is commonly used to measure photospheric velocities [12], and in particular for the study of waves [11]. A line of shorter wavelength would allow smaller files, but would make the velocity measurement less precise.

2.2 Observations

Three sessions were recorded, on 13, 14 and 15 September 2026: 270, 154 and 336 retained scans, over durations of

4.5 h, 2.4 h and 6.0 h. The longer a session, the better neighbouring frequencies can be told apart: the resolution thus improves from 114 microhertz on 14 September to 46 on the 15th. A preliminary session, on 12 September, had served to develop the protocol; it is not used here. Everything that follows is based on the 15 September session alone, the most complete and the best resolved; the other two serve only as a check, to verify that the same ridges are found in them.

That session took place on 15 September 2026, from 07:49 to 13:49 UTC, at Montaigu-Vendée, France. Each scan of the disc is recorded in a separate video file, whose header provides the exact time. Eighteen scans out of 354 were discarded after visual inspection of the reconstructed images, because they showed an obvious defect: wrong geometry, artefacts, or a ripple visible by eye on the disc (Figure 1).

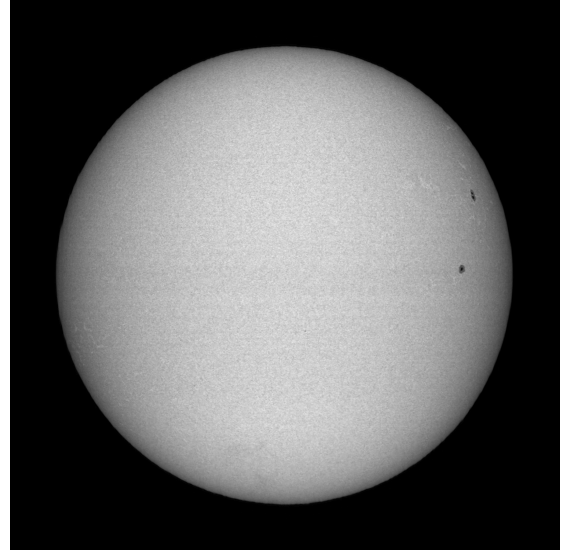


Figure 1: Example of an image reconstructed by JSol'Ex: the solar disc in the iron line, scan of 15 September 2026 at 10:04 UTC. The visual inspection was carried out on images of this kind. The analysed area is the central square, 830 pixels on a side.

Table 2: Observations of 15 September 2026.

Quantity	Value
Scans	336 retained out of 354 recorded
Volume of the retained scans	1.37 TB
Duration	6 h
Interval between scans	58.4 s before the meridian flip, 66.1 s after
Exposure	0.41 to 0.55 ms, adjusted between scans for a constant signal level
Analogue gain	100
Region of interest	3,088 $\times$ 128 px, 16 bits, at 322.6 frames/s
Solar elevation	20° to 46°
Seeing	judged poor

The mount was flipped at the meridian between 12:15 and 12:22 UTC. This flip mirrors the image about one axis; JSol'Ex compensates for it during reconstruction.

¹Prices of the spectroheliograph and the mount read on 19 September 2026 on mlaastro.com; usual retail price of the refractor. Excluding taxes and shipping.

3 Data processing

3.1 Processing carried out in JSol'Ex

Everything is done in JSol'Ex 5.5.0, in a single batch process. Version 5.5.0 is a development version, not yet released at the time of the analysis; the author of this article is the developer of the software. The reference cited is therefore 5.4.2 [13], the latest archived version. The source code is public and 5.5.0 will be archived upon release. For each scan, the software locates the line in the video, reconstructs the images of the disc, fits the outline of the Sun with an ellipse, corrects the tilt and the difference in scale between the two axes of the image, and applies the flip. The analysis itself is a Python script run by the interpreter built into JSol'Ex: it receives the reconstructed images and their acquisition information, then computes the velocities, the diagram and the figures. The scientific analysis requires no other software: reconstructing the images, measuring the velocities and producing the diagram are all done within JSol'Ex. This is what makes it accessible to an amateur.

Acquisition was performed on a mini PC running Windows, with SharpCap and an automatic acquisition script in Python. As that machine is not particularly powerful, the files were processed on a separate computer. Processing the 354 files, including the computation of the diagram, took 2 h 30 on a computer with an AMD Ryzen 9 9950X processor (16 cores, 32 threads) and 128 GB of memory, running Linux Mint 22.3, the files being read from an external SSD connected over USB.

3.2 The processing, step by step

Before going into the details, here is the complete processing, from the shape of the line to the final diagram. The sections that follow take up each step in turn.

1. **Nine images per scan.** JSol'Ex reconstructs the disc at nine neighbouring positions in the line. At every point, these nine values give the shape of the line at that place and at that moment.
2. **Placing the area.** The analysed square is placed by computation, from the position of the disc and the displacement due to the Sun's rotation, whose direction is given by the ephemeris, without being aligned on the content of the images.
3. **Reference shape.** For each point, the mean shape of the line over the 336 scans serves as a reference. It contains everything that does not change at that place, which therefore disappears from the comparison.
4. **Velocity.** The difference between the observed shape and the reference is proportional to the shift being sought. That shift is deduced from all nine points, converted into a velocity, and then the slow drifts are removed, those that do not come from the Sun. They are of two kinds: what affects the whole image at once, such as the instrument warming up, flexure of the setup or the observer's motion relative to the Sun during the day, and what changes slowly at a given point over the six hours. In both cases these are variations far

slower than a five-minute oscillation. We then have one velocity map per scan, that is a film of the surface.

5. **Weighting the scans.** Each scan is given a weight, which depends on its place in the session and on the duration it represents. This is what allows each one to be used at its exact time, despite an irregular cadence.
6. **Sorting by size.** On each map taken separately, the motion is separated according to the size of the ripples: on one side the large waves that span half the area, on the other the finest ripples, and everything in between. Each size also comes in different directions of travel. A Fourier transform is what performs this sorting.
7. **Sorting by rhythm.** Each ripple size is then followed from one scan to the next, over six hours, to find out at what rhythm it comes and goes. A ripple oscillating every five minutes is thus told apart from one oscillating every ten. Here again, it is a Fourier transform, applied this time to time.
8. **The diagram.** In the end we have one number for each size and each rhythm: the power of the corresponding motion. Since the Sun favours no particular direction, all directions of the same size are averaged together, which reduces the noise. All that remains is to display these numbers, size on the horizontal axis and frequency on the vertical one: that is the diagram. A resonance mode is precisely a regular ripple that oscillates regularly: it therefore ends up concentrated at one point of the diagram instead of spreading across it, and that is why ridges appear.

3.3 Images and analysed area

For each scan, JSol'Ex reconstructs not one but nine images, taken at nine neighbouring positions in the line: four on one side of the line centre, the centre itself, and four on the other side. These nine images give, at every point of the disc, the shape of the line.

The analysis covers an 830×830 pixel square centred on the disc, that is 417,000 km on a side. This conversion to kilometres uses the scale at disc centre; it is an approximation, acceptable here because the area stays close to the centre, where the surface is seen face-on. This choice follows the work of Gizon and Birch [9], who show that an area centred on the disc is enough to obtain a good diagram.

The disc radius, which JSol'Ex fits on each image, is not a measurement of the Sun's diameter: it serves as a ruler to convert pixels into kilometres, and thereby sets the scale of wave sizes. The analysis itself covers only the central square.

Over six hours, the rotation of the Sun moves the surface far enough that it must be taken into account: otherwise, different regions would be compared from one scan to the next. As the granulation is not resolved and the centre of the disc shows no stable detail, the area cannot be aligned on the images themselves. It is therefore tracked by computation: JSol'Ex provides the position of the disc in each image, and the displacement due to rotation is computed with the law of Snodgrass and Ulrich [14]. The

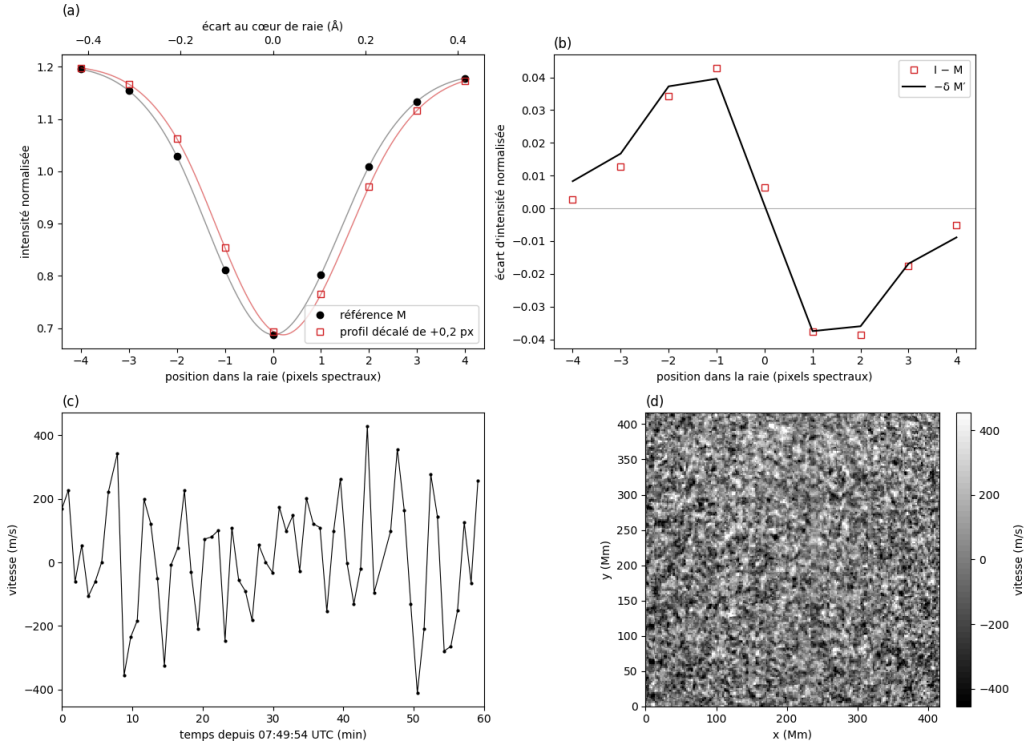


Figure 2: Doppler velocity measurement, 15 September 2026. (a) Mean shape of the line and the same shape slightly shifted. (b) Difference between the two, which is used to measure the shift. (c) Mean velocity of a small square at the centre of the area during the first hour: the five-minute oscillation can be seen directly. (d) Velocity map of one scan.

direction of that displacement comes from the ephemeris, which gives the tilt of the Sun's rotation axis in the image on the date of the observation.

3.4 Checking that the area follows the surface

This tracking has to be exact, as experience showed. If the square does not follow the surface, the surface slowly slips across the analysed area. A wave travelling in the direction of the slip then seems to beat slightly faster, and a wave travelling the other way slightly slower. On the diagram, every ridge splits in two, all the more so as the waves are small. A method for measuring the currents beneath the surface of the Sun actually relies on this effect [9].

A first version of this work had this defect. The direction of the displacement was measured on the images, and a flaw of the instrument biased it. The surface slipped at 1.5 km/s across the area. The ridges were split, and one of the two halves of the f mode, shifted downwards, looked like an unknown band that we had been unable to explain.

A check now verifies the tracking. Velocity maps averaged over twenty minutes wipe out the oscillations and keep only the long-lived patterns of the surface. Comparing these maps from one block to the next gives the remaining slip: 66 m/s here, half a pixel per hour, whereas the Sun's rotation moves the surface at 1.9 km/s. Such a remainder has no visible effect on the diagram.

3.5 Measuring the velocities

The principle is to compare, at each point, the shape of the line observed at a given moment with its mean shape over the whole session (Figure 2).

The shape of the line. The nine images give nine brightness measurements that trace the dip of the line. They are divided by their mean, which removes brightness variations due to sky transparency, exposure time or the darkening of the disc towards its edge. Only the shape remains.

The reference. For each point of the disc, the mean shape of the line over the 336 scans is computed. This mean contains everything that does not change during the session, such as the defects specific to that point of the image. By comparing each measurement with the reference of the same point, these defects cancel out by themselves, without having to be modelled.

The shift. A motion of the surface shifts the line in wavelength. This shift is very small: a velocity of 300 m/s moves the line by only one twentieth of a pixel. It therefore cannot be read directly, but it can be deduced from the difference between the observed line and the reference line. This difference is proportional to the shift being sought, and its measurement uses all nine points at once, which makes it much more precise than a single pixel (Figure 2 b). The shift is then converted into a velocity.

There remain the slow drifts, which do not come from the Sun. For each scan, the median value of the whole image is removed, which eliminates anything affecting the whole image at once. For each point, its mean value and a straight line fitted by least squares over the six hours are then removed. Removing a single straight line over the whole series only affects variations slower than one oscillation per session, far below the frequencies studied here. The measurement noise is 212 m/s for one pixel and

one scan; it is estimated from the scatter from one scan to the next, that is the standard deviation of the difference between two consecutive velocity maps, scaled to a single map. Averaging over 5×5 pixel squares, it drops to 42 m/s, whereas the measured velocity still varies by 157 m/s, nearly four times the noise: the variation is therefore clearly detectable in these data (Figure 2 c). Its solar origin, however, cannot be deduced from this comparison alone; it rests on the results that follow.

3.6 From the velocity cube to the diagram

At this stage, we have 336 velocity maps, one per scan: a film of the surface, taken every minute for six hours. These motions now have to be sorted by size and by rhythm.

Sorting by size is done on each map separately. It is a decomposition into ripples: we look for how much of the motion takes the form of large ripples, and how much takes the form of small ones. The calculation used is the Fourier transform, a standard tool that breaks an image down into ripples of different sizes and orientations.

Sorting by rhythm comes next, by following each ripple size from one scan to the next. A difficulty arises here: the scans are not evenly spaced in time, and the interval even changes during the session. Rather than forcing them onto a regular schedule, which would amount to giving them a false time, each scan is used at the exact time it was taken. Scans at the beginning and end of the session count slightly less than those in the middle, which avoids creating artefacts at the ends of the series, and each one is weighted according to the duration it represents.

The result is the diagram we are looking for: for each ripple size and each rhythm, the power of the corresponding motion. In practice, each scan is projected onto sine waves of increasing frequency, evaluated at its actual time and given its weight: this is a weighted Fourier transform, computed at the observation times rather than on a regular grid. The frequencies examined are multiples of the inverse of the session duration, that is multiples of 46 microhertz: this is the frequency resolution of the series. At the other end, the interval of about one minute between scans sets the highest accessible frequency, known as the Nyquist frequency: 8.6 millihertz before the meridian flip, 7.6 after. The background level, which we measure between 6 and 8 millihertz, comes close to it, which calls for caution near 8. Finally, since the waves travel in all directions, the powers are averaged over all orientations of the same ripple size.

A word about the uncertainties, given everywhere at one standard deviation. The waves of the Sun are excited at random: the power measured in one cell of the diagram therefore fluctuates a great deal from one cell to the next, and this fluctuation sets the uncertainty of each mean. The computation takes into account the fact that neighbouring cells are not independent, which is a consequence of the way each map is prepared before the Fourier transform. For the frequency of the ridges, the uncertainty is obtained differently: the waves are sorted into eight groups according to their direction of propagation, the ridge is

measured in each group, and the scatter of these eight measurements gives the uncertainty. These uncertainties are statistical: they say nothing about the flaws of the instrument, which are discussed in the limitations.

A word about this power, since it is the quantity shown in every figure. It is the square of a velocity, therefore a way of measuring the strength of the motions, wave size by wave size and frequency by frequency. Why this quantity rather than the velocity itself? Because the oscillations are sustained at random by convection: at any given moment some regions rise while others fall, and a simple average of the velocities would cancel out. By squaring, the information is kept whatever the sign. The scale, on the other hand, is arbitrary: it depends on computational conventions, which does not matter here, since we only compare ratios between two frequency ranges.

4 Results

The figures and values in this section come from the processing described above, with no further correction. Section 5 presents the gain obtained by correcting the mount's tracking defect.

4.1 The diagram

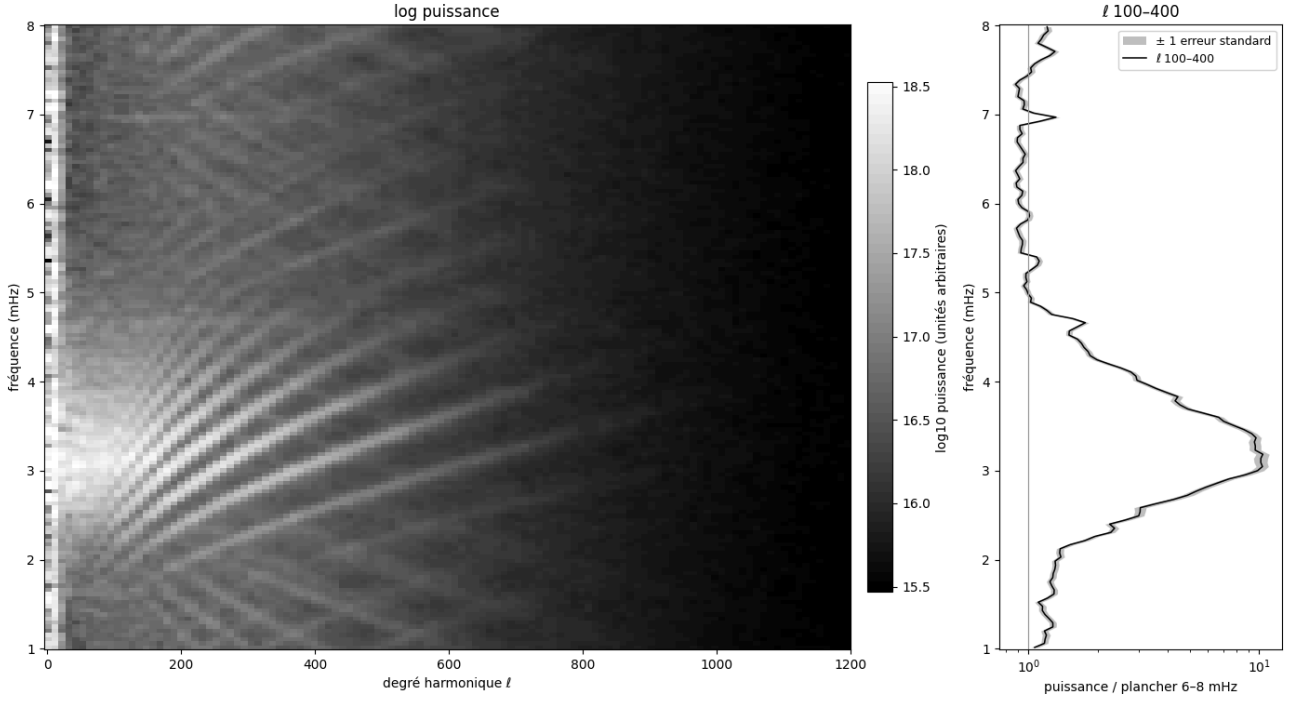
Figure 3 shows the diagram obtained, and Figure 4 the same diagram with reference marks added.

The diagram clearly shows what Ulrich predicted: the power is not distributed at random, but concentrated along lines. It peaks near 3.2 mHz, that is one oscillation every 5 minutes and 13 seconds; the top is flat, and the exact position of the maximum carries little meaning. Between 2.5 and 4 mHz, it is nearly seven times stronger than the background level measured between 6 and 8 mHz.

Above 5 mHz, the power drops and reaches a floor around 5.5 mHz. This value is of the order of the acoustic cut-off frequency of the photosphere: beyond it, the waves propagate more readily into the upper layers instead of remaining confined in the cavity where they resonate.

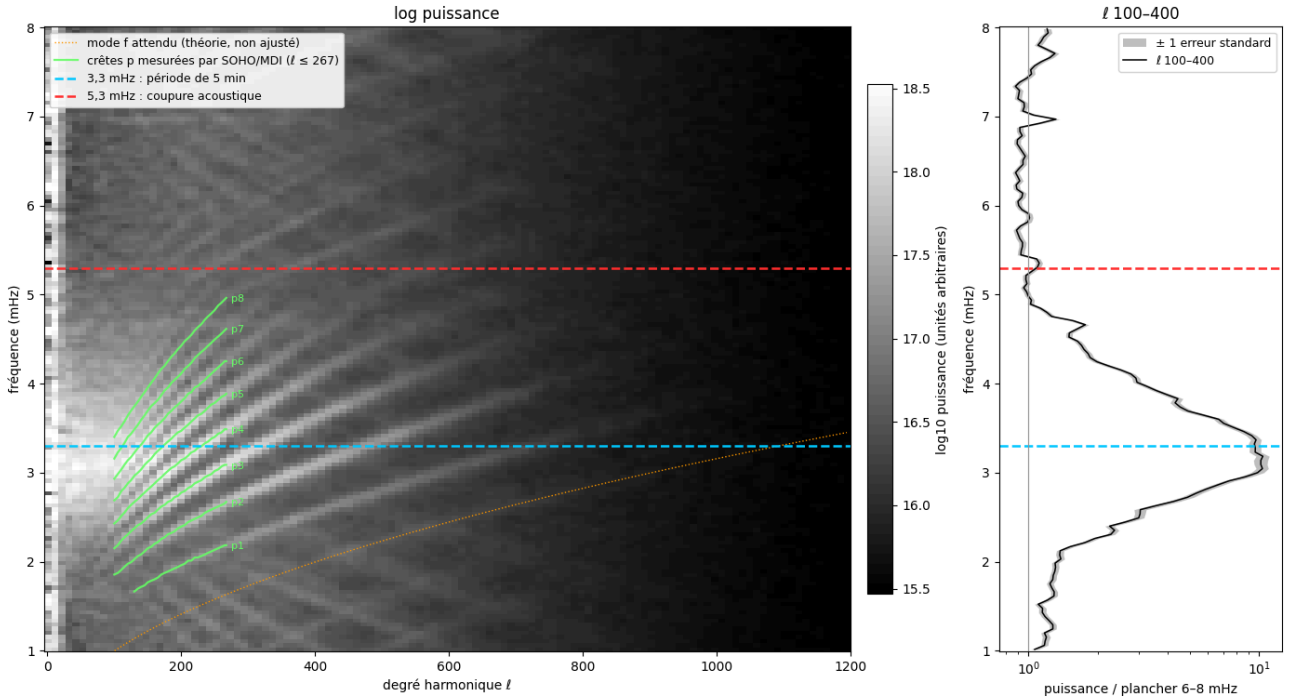
4.2 The ridges compared with space measurements

Separate ridges are visible from degree 100 to about degree 900. The frequency of each ridge is not simply read at the position of the maximum: a curve is fitted to the five points surrounding that maximum, which locates the peak more finely than a single point of the spectrum. These values are compared with the ridges of a published diagram from the SOHO satellite (Table 3). The uncertainty given concerns only our measurement; reading the SOHO ridges from a published figure has its own imprecision, of a few tens of microhertz.



Cédric Champeau · 15 septembre 2026, 07:49-13:49 UTC · MIAstro SHG 700 · SVBony SV48P 102, 102/663 mm · G3M678M · raie 5434,53 Å

Figure 3: The diagram: the power of the motions as a function of degree, on the horizontal axis, and frequency, on the vertical axis. The brighter an area, the stronger the corresponding motion. On the right, the mean power for degrees 100 to 400, with its uncertainty.

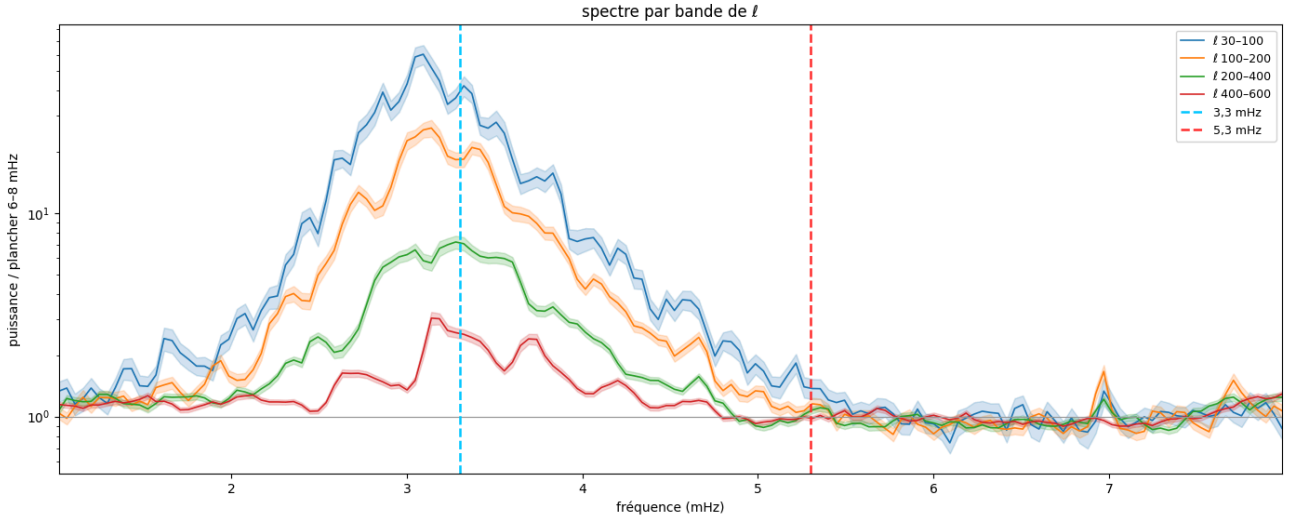


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Figure 4: Same diagram with reference marks. In orange, the theoretical position of the f mode, computed without any adjustment from the surface gravity wave relation given by Gizon and Birch [9], with the gravity at the Sun's surface and its radius as the only constants. In green, the ridges read from a published diagram from the SOHO satellite [15], built from the measurements of MDI's medium- ℓ programme [16]. The straight lines mark 3.3 and 5.3 mHz.

Table 3: Frequencies of the six ridges, in millihertz, with their uncertainty, compared with the values read from the SOHO diagram.

Ridge	Degree 199	SOHO	Degree 252	SOHO
p1	1.941 ± 0.015	1.96	2.124 ± 0.014	2.14
p2	2.392 ± 0.006	2.39	2.619 ± 0.005	2.61
p3	2.761 ± 0.006	2.77	3.015 ± 0.005	3.03
p4	3.128 ± 0.007	3.14	3.406 ± 0.005	3.41
p5	3.460 ± 0.007	3.47	3.787 ± 0.003	3.79
p6	3.795 ± 0.008	3.79	4.148 ± 0.006	4.16



Cédric Champeau · 15 septembre 2026, 07:49-13:49 UTC · MIAstro SHG 700 · SVBony SV48P 102, 102/663 mm · G3M678M · raie 5434,53 Å

Figure 5: Mean power as a function of frequency, for four ranges of degree. The shaded areas represent the uncertainty; it is often thinner than the line itself.

This agreement is the most robust result of this work. The differences range from -0.019 to $+0.009$ mHz: all twelve ridges fall within 20 microhertz of the SOHO value, less than half a resolution element, and no difference is significant given the uncertainties. The largest differences are those of the p1 ridge, whose peak is the least sharp. The agreement relies on no tuning at all: the frequency scale comes solely from the time stamps of the files, and the size scale solely from the disc radius measured in pixels. Two shorter sessions, on 13 and 14 September, show the same ridges, with lower precision owing to their shorter duration.

4.3 Comparison with published frequencies

The previous comparison relies on reading a figure. The MDI project also publishes its frequencies as lists of numbers, fitted directly to its spectra [17]. Unfortunately, these lists do not extend far enough towards small waves to cover degrees 199 and 252: at degree 199 they contain only the first ridge, and at degree 252 none at all. The comparison is therefore repeated at degree 126, where all six ridges are tabulated (Table 4).

Table 4: Our six ridges at degree 126, compared with the frequencies published by MDI. Frequencies are in millihertz, differences in microhertz.

Ridge	Published value	Our measurement	Difference
p1	1.6441 ± 0.0003	1.653 ± 0.020	+9
p2	2.0072 ± 0.0008	2.023 ± 0.022	+16
p3	2.3404 ± 0.0014	2.329 ± 0.009	-12
p4	2.6356 ± 0.0024	2.636 ± 0.007	0
p5	2.9201 ± 0.0038	2.926 ± 0.008	+6
p6	3.1902 ± 0.0041	3.181 ± 0.008	-9

The mean difference is 9 microhertz, a fifth of the resolution of our series, and each difference is smaller than one and a half times the uncertainty of the measurement. The same comparison gives 22 at degree 147 and at degree 168 alike, the poorly defined p1 ridge accounting for most of the difference. At degree 147 and below, however, almost

all our frequencies are slightly too high, by 37 microhertz on average at degree 105: this is more than the uncertainty, and we do not know its cause.

4.4 The smallest waves fade away

Figure 5 compares the power of the oscillations for four ranges of size.

Table 5: The contrast of the oscillations drops as the waves get smaller.

Degree range	Power of the oscillations, relative to the background
30-100	27.8 ± 1.1
100-200	13.8 ± 0.4
200-400	4.78 ± 0.09
400-600	1.92 ± 0.03

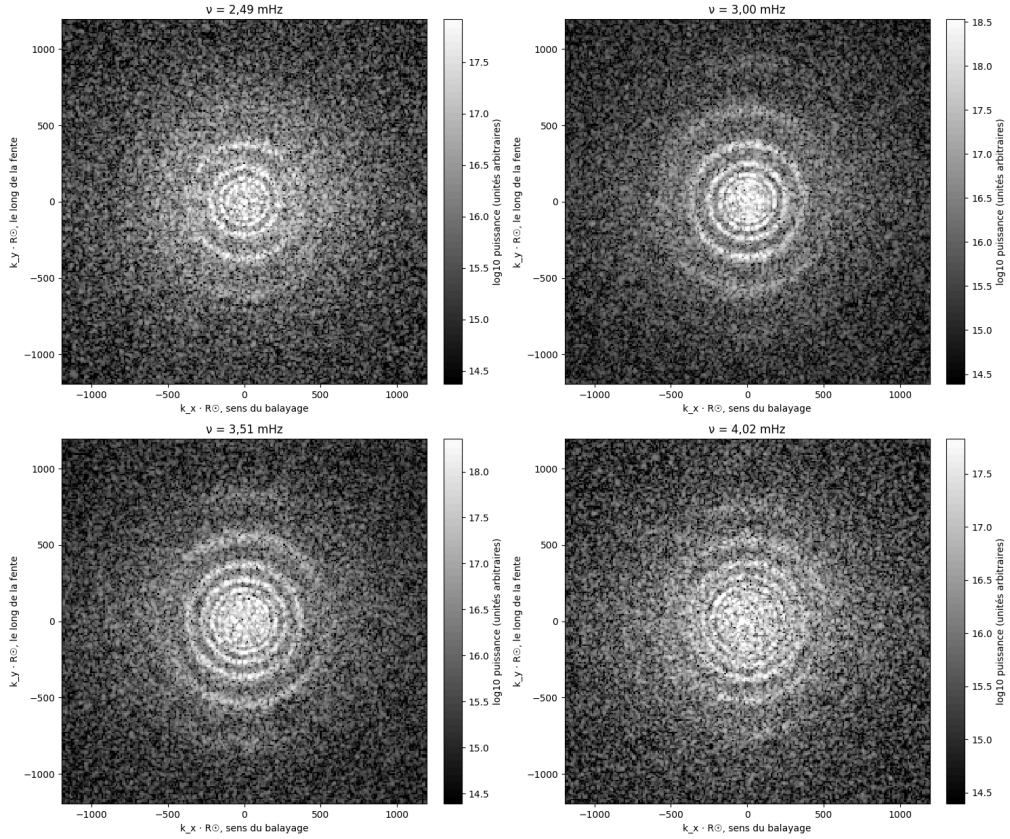
The power of the oscillations, relative to the background, is about fifteen times weaker for degrees 400 to 600 than for degrees 30 to 100. Two causes add up, and these data cannot separate them: turbulence in the Earth's atmosphere, which blurs the finest details, and the resolving power of the instrument, which attenuates them. This is the main limitation of an observation made from the ground, and it explains why degree 800 is the practical detection limit for the ridges here.

4.5 Rings rather than ridges

If we choose one frequency and look at all directions at once, each ridge becomes a ring (Figure 6). The radius of the ring gives the size of the waves, and the concentric rings correspond to the successive families. They are sharp between 2.5 and 3.5 mHz, and more diffuse at 4 mHz.

4.6 Sharpness depends on direction

Between 2.5 and 4 mHz, the power is nearly seven times stronger than the background, on average over all directions. But this average hides a difference: for waves travelling along the scan direction the ratio is only 4.5, whereas it reaches 10.9 for those travelling along the slit. The instrument therefore sees less well along the scan



Cédric Champeau · 15 septembre 2026, 07:49-13:49 UTC · MIAstro SHG 700 · SVBony SV48P 102, 102/663 mm · G3M678M · raie 5434,53 Å

Figure 6: View of the diagram at four fixed frequencies. Each image shows the power according to the direction and size of the waves; the centre corresponds to the largest ripples. The horizontal axis is the scan direction, the vertical axis that of the slit.

direction, which is also the one in which the imperfection of the mount described in Section 5 acts. This difference is far larger than the statistical uncertainty: the mean values given here depend on the instrument as much as on the Sun.

4.7 The f mode

There is another family of waves, called the f mode. These are not sound waves but surface waves, comparable to waves on water. Their theoretical position is plotted in orange in Figure 4. A faint ridge follows it beyond degree 450: it is examined in Section 6.

5 An imperfection of the mount, measured and corrected

5.1 The origin of the problem

The mount drives the telescope to compensate for the rotation of the Earth, but it does so with a small irregularity that repeats every 430 seconds, with an amplitude of about 16 arcseconds. The manufacturer confirmed this period and amplitude.

While a scan is being recorded, this irregularity slightly changes the speed at which the image of the Sun drifts across the slit. JSol'Ex corrects the mean scale of each scan and recentres the disc, but a distortion remains that shifts the content of the image by a few pixels, differently from one scan to the next. Successive velocity maps therefore

end up slightly offset from one another, which blurs the ridges.

5.2 The correction

It is applied to the velocity maps already computed, without reprocessing the videos, and takes a few minutes. It proceeds in three steps.

1. **Measure.** Each velocity map is compared with the previous one to determine how far its content has slipped. The measurement is made in four strips spread along the scan, since the slip is not the same everywhere. Velocity maps are well suited to this measurement because they are rich in detail, unlike ordinary images of the disc.
2. **Model.** From these measurements, only what repeats at the mount's own rhythms is kept: 431.6 seconds and two faster rhythms derived from it. Everything else is ignored, which avoids correcting anything other than the mount. The model accounts for 29 to 48 % of the observed slip depending on the strip, and the slip reconstructed in this way reaches 6 pixels before the meridian flip, and 13 pixels after (Figure 7 a).
3. **Correct.** Each velocity map is put back in place according to this model, and the diagram is recomputed.

To make sure that the gain really comes from the mount and not from chance, the same correction was applied to randomly drawn scans. If the method improved things

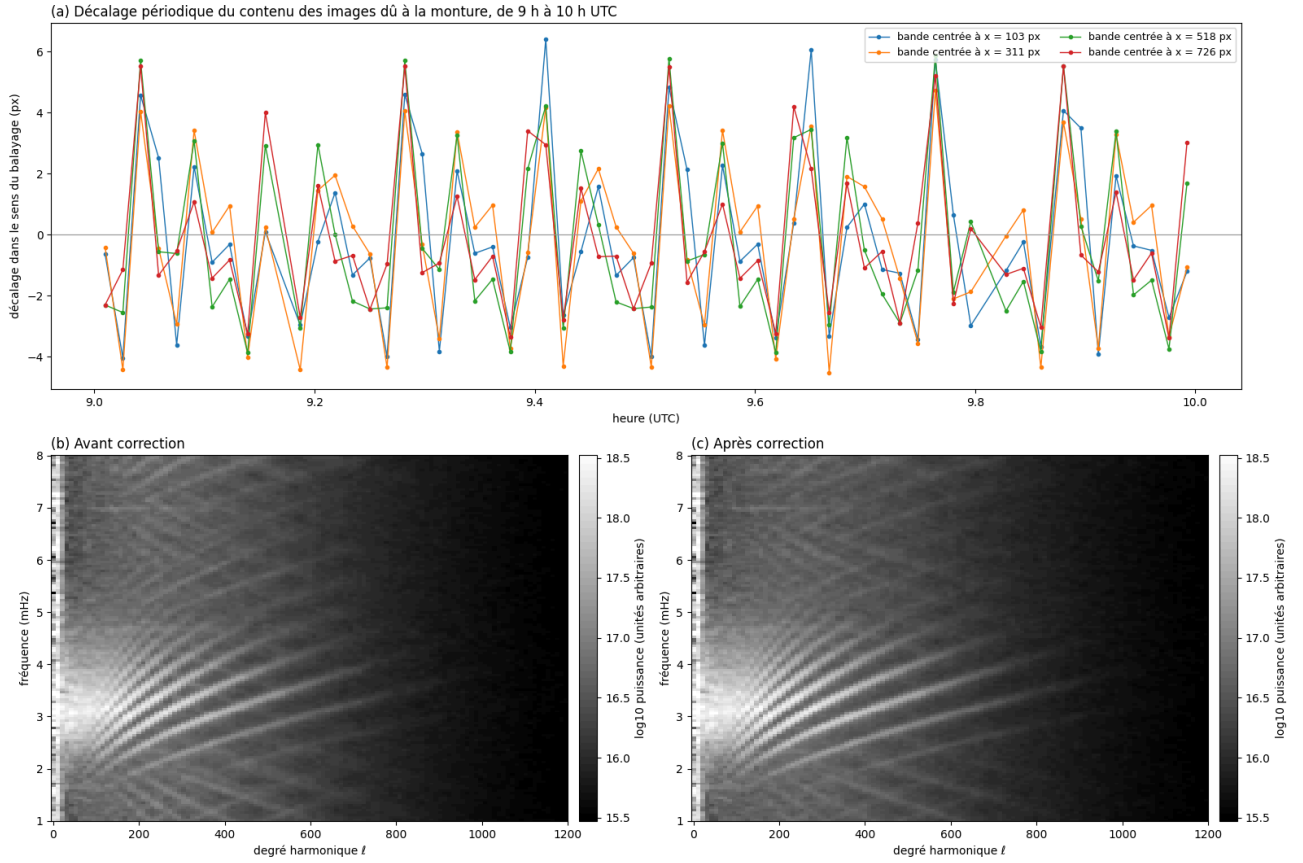


Figure 7: Correction of the mount's irregularity. (a) Reconstructed slip of the image content, for the four strips, between 9 and 10 h UTC. (b) Diagram before correction. (c) After correction, with the same grey scale.

Table 7: The same measures before and after correction.

Measure	Without correction	After correction
Frequency of the maximum	3.19 ± 0.05 mHz	3.05 ± 0.05 mHz
Power of the oscillations, degrees 100 to 400	6.77 ± 0.11	9.62 ± 0.16
— along the scan direction	4.5	7.2
— along the slit	10.9	13.5
Power just below the trapping limit	1.86 ± 0.03	2.31 ± 0.03
Contrast, degrees 30 to 100	27.8 ± 1.1	32.9 ± 1.3
Contrast, degrees 100 to 200	13.8 ± 0.4	19.1 ± 0.5
Contrast, degrees 200 to 400	4.78 ± 0.09	7.02 ± 0.14
Contrast, degrees 400 to 600	1.92 ± 0.03	2.57 ± 0.04
Difference of the ridges from the SOHO values	-0.019 to $+0.009$ mHz	-0.016 to $+0.008$ mHz

merely by construction, this control would improve as well.

5.3 The result

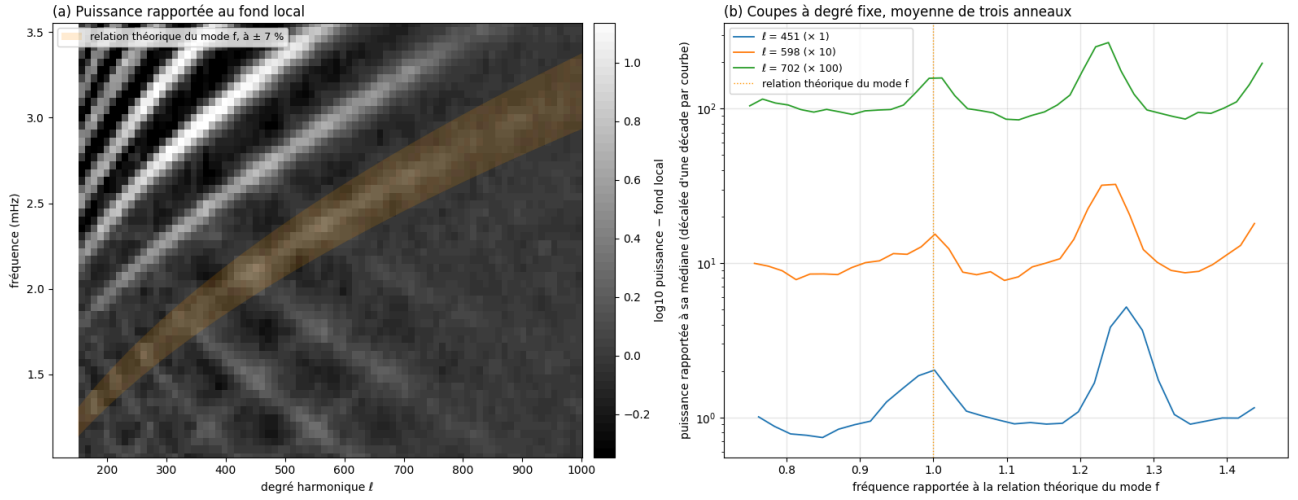
Table 6: Effect of the correction, compared with the randomly drawn control.

Measure	Before	After	Control
Power along the SOHO ridges, relative to the background	0.344	0.368	0.323
Sharpness of the ridges, degrees 200 to 600	16.8	18.3	13.2
Sharpness of the ridges, degrees 600 to 800	5.8	7.1	4.1
Power of the oscillations, degrees 100 to 400	6.8	9.7	5.6

The correction improves every measure, whereas the randomly drawn control degrades them (Table 6). This result is consistent with a gain due to correcting the periodic motion of the mount. It mainly affects the waves travelling along the scan direction, which is precisely the direction in which the mount acts. On the diagram, oblique patterns with no physical meaning fade away (Figure 7 b and c).

Wind and turbulence in the atmosphere also move the images from one scan to the next, but irregularly. Only the regular part of the motion, that of the mount, is corrected here. The rest remains: it makes the ridges a little less sharp, without changing their frequency.

The contrast of the oscillations increases by half (Table 7), while the frequencies of the ridges move by no more than the frequency resolution: the correction makes



Cédric Champeau · 15 septembre 2026, 07:49-13:49 UTC · MIAstro SHG 700 · raie 5434,53 Å · erreur de monture corrigée

Figure 8: The f-mode region, after correction of the mount. (a) The power after subtraction of the background, which brings out the fine structures. The orange area surrounds the theoretical position of the f mode, within $\pm 7\%$. (b) Cuts of the diagram at three degrees: the power as a function of frequency, divided by the theoretical frequency of the f mode. The peak at 1 is the f mode, the one near 1.24 is the p1 ridge.

the diagram sharper without displacing what it contains. The gain is found in both directions, but the difference between them remains (Section 4.6).

6 The f mode

6.1 Detection

Figure 8 enlarges the f-mode region, after correction of the mount. A ridge follows the theoretical curve there from about degree 450 to degree 900. Its frequency is measured like that of the other ridges (Table 8). The contrast tells by how much the peak rises above the surrounding background: at 1, there is nothing to see.

Table 8: Frequency of the f-mode ridge, in mHz, compared with theory.

Degree	Theory	Measured	Difference	Contrast
451	2.120	2.104 $\pm$ 0.009	-0.7 %	2.6
503	2.240	2.251 $\pm$ 0.013	+0.5 %	1.5
545	2.331	2.347 $\pm$ 0.006	+0.7 %	2.0
598	2.441	2.435 $\pm$ 0.015	-0.2 %	1.9
650	2.546	2.545 $\pm$ 0.010	0.0 %	1.6
702	2.646	2.656 $\pm$ 0.010	+0.4 %	1.7
755	2.743	2.728 $\pm$ 0.013	-0.6 %	1.5
797	2.819	2.844 $\pm$ 0.030	+0.9 %	1.2
849	2.910	2.918 $\pm$ 0.034	+0.3 %	1.1
902	2.998	2.997 $\pm$ 0.008	-0.1 %	1.4

From degree 451 to degree 902, the ridge stays within 1 % of theory, which is of the order of the uncertainty of our measurements. Published measurements are far finer: the MDI frequencies [17] place the f mode 0.7 to 1.2 % above the theoretical curve up to degree 300, and other measurements find it 13 microhertz from theory at degree 880 [18]. Its variations over the solar activity cycle are smaller still, a few hundredths of a microhertz [19]. Our data cannot show such differences. They show that the ridge is there, at the expected position.

Below degree 400, the f mode is not detected: the contrast falls to about 1 and the peak is no longer defined. Its

power is low there, and traces of the imperfection of the mount, which run down at an angle in the figure, cross that region (Figure 8 a).

6.2 A check: the mean density of the Sun

The mass and radius of the Sun are known accurately. The f mode makes it possible to recover their ratio from the diagram alone, which makes it an overall check of the measurement, independent of the comparison with SOHO.

For a surface wave, the rhythm depends only on gravity and on the length of the wave: $\omega^2 = g \cdot k$, where ω is the angular frequency (2π times the frequency) and k the wavenumber, which is ℓ / R for a degree ℓ on a sphere of radius R . The gravity at the surface of a body is $g = G \cdot M / R^2$. Substituting, the radius survives only in the ratio M / R^3 , which is nothing but the mean density up to a factor:

$$\bar{\rho} = \frac{3\omega^2}{4\pi G \ell}$$

Each frequency of Table 8 therefore gives a density. At degree 650, a frequency of 2.545 mHz gives 1,406 kg/m³. Neither the distance of the Sun nor its radius in kilometres is involved: the degree is measured in pixels, the frequency in seconds.

Table 9: Mean density of the Sun derived from each f-mode frequency.

Degree	Frequency (mHz)	Density (kg/m ³)
451	2.104 $\pm$ 0.009	1,386 $\pm$ 12
503	2.251 $\pm$ 0.013	1,421 $\pm$ 16
545	2.347 $\pm$ 0.006	1,426 $\pm$ 7
598	2.435 $\pm$ 0.015	1,400 $\pm$ 17
650	2.545 $\pm$ 0.010	1,406 $\pm$ 11
702	2.656 $\pm$ 0.010	1,417 $\pm$ 11
755	2.728 $\pm$ 0.013	1,391 $\pm$ 13
797	2.844 $\pm$ 0.030	1,433 $\pm$ 30
849	2.918 $\pm$ 0.034	1,415 $\pm$ 33
902	2.997 $\pm$ 0.008	1,406 $\pm$ 8

The mean of the ten values, weighted by their uncertainties, is $1,410 \pm 4 \text{ kg/m}^3$, for an accepted value of $1,408 \text{ kg/m}^3$ (Table 9). This check verifies both scales of the diagram at once: since the density is proportional to ω^2 / ℓ , a 1 % error in the frequencies would shift it by 2 %, and a 1 % error in the degrees by 1 %. The surface-wave formula is not quite exact on the real Sun: the frequencies published by MDI [17] depart from it by 0.7 to 1.2 % between degrees 150 and 300, and by less than 0.5 % at degree 880 [18]. The expected agreement on the density is therefore of order 1 to 2 %, and the agreement obtained, 0.2 %, is partly fortuitous. With the tabulated radius, 695,700 km, this density corresponds to a surface gravity of 274.2 m/s^2 and a mass of $1.989 \times 10^{30} \text{ kg}$, for accepted values of 274.0 m/s^2 and $1.988 \times 10^{30} \text{ kg}$.

7 Discussion

7.1 Comparison with ground-based observations

Deubner [3] could distinguish three or four ridges from the ground; the diagram presented here separates six of them, and allows them to be identified by comparison with SOHO.

That the f mode is far weaker than the other ridges is not surprising. Gizon and Birch [9] compare two diagrams obtained at the same time, one from space with SOHO, the other from the ground with the TON network, in conditions close to those of this work: the f-mode ridge is markedly weaker as seen from the ground. With a 102 mm refractor and a poor sky, its low contrast is consistent with what published ground-based observations show.

7.2 Limitations

- **Atmospheric turbulence.** The f mode stands out mostly among small waves, which is precisely where turbulence erases the most power.
- **Direction.** The power of the oscillations, relative to the background, is twice as low along the scan direction as along the slit (Section 4.6).
- **Tracking of the surface.** The remaining slip, 66 m/s, is too small to split the ridges visibly. The check relies on patterns of the surface whose trace, at the centre of the disc, is faint.
- **The correction of the mount.** It only deals with the slip along the scan direction, with three rhythms and four strips. The slip is measured on the very data that are then corrected; the control shows that the gain does come from the mount, but it cannot be ruled out that a small share of solar signal at the same rhythms was affected.
- **The reference values at degrees 199 and 252.** They were read from a published figure, for lack of frequency lists covering these degrees: the position of the ridges is located on the image and converted using the axis graduations. The accuracy of such a reading is of the same order as the measured differences, so it is not possible to tell what comes from the Sun, from the instrument, or from the reading. Comparison with the published lists is only possible at degree 168 and below.

- **Wind and turbulence.** They too shift the content of the images from one scan to the next, but irregularly. Only the part that repeats, due to the mount, is corrected; the rest remains, weakens the contrast of the ridges and may create bands of power, without displacing the frequencies.
- **A single session.** The available sessions have not been combined.

8 Conclusions and outlook

We have not identified any equivalent velocity measurement made with an amateur instrument, after searching bibliographic databases, preprint archives, and the websites and forums of the amateur spectroheliograph community; from the brightness of the surface alone, a diagram of this kind has already been obtained [8]. The Sun's five-minute oscillations are detected here in velocity with an amateur spectroheliograph, and their families of waves separated one by one. Six ridges are resolved up to degree 900, and their frequencies are compatible, within the accuracy allowed by reading the published figure, with those of the SOHO diagram. JSol'Ex's batch processing and its Python interpreter make this measurement accessible.

The f mode is detected from degree 450 to degree 900, at the position predicted by theory, and its frequency recovers the mean density of the Sun, hence its mass, which confirms that the diagram is correct. One lesson, finally: one must check that the analysed area really follows the surface before interpreting the diagram, otherwise split ridges can be mistaken for new features.

Several avenues remain open: integrating the mount correction into JSol'Ex processing, combining several sessions to gain precision, and extending the comparison with published tables towards high degrees. Beyond degree 190, the reference frequencies are no longer given by the observatories' usual tables, but by specialised analyses.

9 Acknowledgements and AI use statement

This article was written with the assistance of the artificial intelligence Claude (Anthropic). The measurement protocol and the processing pipeline were designed by the author from the scientific literature. The software implementation of that pipeline was carried out in cooperation with the AI: it produced the Python code, while the author directed its design, identified numerous errors and had them corrected, and validated the final version. Likewise, the analysis of the results was written with the assistance of the AI, which proposed interpretations and comparisons with the literature. The author reviewed and validated the final text.

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